

SMART INDIA HACKATHON 2025



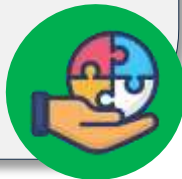
TITLE PAGE

- **Problem Statement ID** – 25016
- **Problem Statement Title** - Automated Student Attendance Monitoring and Analytics System for Colleges
- **Theme** - Smart Education
- **PS Category** - Software
- **Team ID** - 82181
- **Team Name** –The Innovexa



Problem Resolution

- Combining **multi-factor authentication** (biometric + dynamic QR code).
- Monitoring real-time** engagement in online classes through session duration and polls.
- Providing teachers with a controlled **override option** in case of technical failure.



IDEA TITLE

Idea/Solution

For online sessions:

- Attendance is marked if the student stays active for **at least 85%** of the session & requires the student to answer **interactive polls** during class.

For offline sessions:

- The teacher displays three QR codes one by one (each valid for 3 seconds), which students complete **biometric verification & scan** using their mobile device to mark attendance.



Unique Value Proposition

- Tamper-proof attendance:** **Dynamic QR** codes with expiry time + biometric authentication
- Engagement-based validation:** Attendance linked not just to presence but also to participation.
- Teacher override mechanism:** No genuine student is marked absent due to technical glitches.





Frontend: React

Responsive web interface built with React.



Backend: Nodejs

Business logic and attendance verification workflows.



Database: MongoDB

Ideal for handling large and changing datasets efficiently.



Biometric: OpenCV

Face recognition or fingerprint scanners.



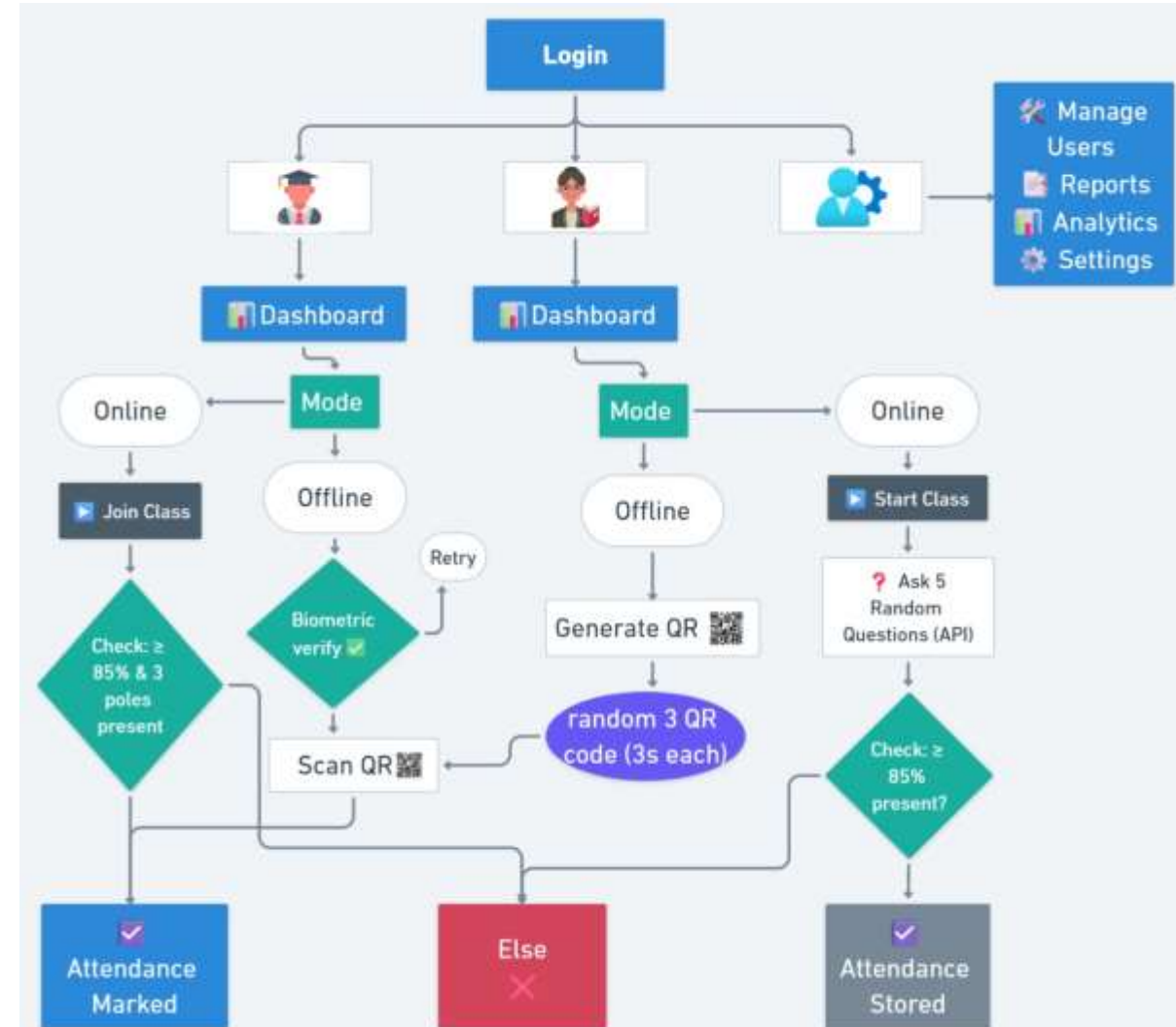
Analytics: Power BI

Data visualization using dashboards



QR : qrcode.react

Randomized, time-limited QR generation



Technological Feasibility:

Leverages **existing tech** (cameras, fingerprints, QR code scanners) with integration into a **single Platform**.



Economic Viability:

Cost-effective since it uses already available resources in colleges (Smartboards and Smartphones).



Operational Feasibility:

Simple for **students & faculty** - scanning a QR code or participating in polls is **minimal effort**.



Scalability:

Enable institutions to handle **thousands of students** and a high volume of **daily attendance** events.



Ensures **fair and transparent attendance monitoring** for students. 01

Minimizes **proxy attendance and misuse** of the system. 02

Encourages the **active student engagement** in classes. 03

Reduces the **manual effort and workload** for faculty. 04

Provides the **real-time attendance tracking** for accuracy. 05

Helps institutions with **streamlined processes & analytics**. 06



01 Eliminates **manual errors** in attendance records.

02 Prevents **fraud and misuse** in attendance management.

03 Saves **time and effort** for faculty and staff.

04 Provides **data-driven insights** for academic support.

05 Builds a **transparent system** between students and teachers.

06 Enhances **overall efficiency** in attendance management.

- ❖ **Biometric Attendance Systems:** “Enhancing Biometric Attendance Systems for Educational Institutions,” by **Vedant Mankar**, IJISRT .
- ❖ **QR Code-Based Attendance Systems:** “QR Code Based Attendance Management System,” by **K.Narangamma** RSIS International.
- ❖ **Engagement Validation in Online Classes:** “Automated Attendance System” by **Krish Makkar**, SSRN.